

Fall 1982

Problem 1 (Fa82). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous nowhere vanishing function, and consider the differential equation

$$\frac{dy}{dx} = f(y)$$

1. For each real number c , show that this equation has a unique, continuously differentiable solution $y = y(x)$ on a neighborhood of 0 which satisfies the initial condition $y(0) = c$.
2. Deduce the conditions on f under which the solution y exists for all $x \in \mathbb{R}$, for every initial value c .

Problem 2 (Fa79, Su80, Fa82). Consider the polynomial ring $\mathbb{Z}[x]$ and the ideal $\mathfrak{J}$ generated by 7 and $x - 3$.

1. Show that for each $r \in \mathbb{Z}[x]$, there is an integer α satisfying $0 \leq \alpha \leq 6$ such that $r - \alpha \in \mathfrak{J}$.
2. Find α in the special case $r = x^{250} + 15x^{14} + x^2 + 5$.

Problem 3 (Fa82). Let $\cot \pi z = \sum_{n=-\infty}^{\infty} a_n z^n$ be the Laurent expansion for $\cot \pi z$ on the annulus $1 < |z| < 2$. Compute the a_n for $n < 0$.

Problem 4 (Sp78, Fa82, Fa86). Let M be a matrix with entries in a field $\mathbb{F}$. The *row rank* of M over $\mathbb{F}$ is the maximal number of rows which are linearly independent (as vectors) over $\mathbb{F}$. The *column rank* is similarly defined using columns instead of rows.

1. Prove row rank = column rank.
2. Find a maximal linearly independent set of columns of

$$\begin{pmatrix} 1 & 0 & 3 & -2 \\ 2 & 1 & 2 & 0 \\ 0 & 1 & -4 & 4 \\ 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \end{pmatrix}$$

taking $\mathbb{F} = \mathbb{R}$.

3. If $\mathbb{F}$ is a subfield of $\mathbb{K}$, and M has entries in $\mathbb{F}$, how is the row rank of M over $\mathbb{F}$ related to the row rank of M over $\mathbb{K}$?

Problem 5 (Fa82). Let $\varphi_1, \varphi_2, \dots, \varphi_n, \dots$ be nonnegative continuous functions on $[0, 1]$ such that the limit

$$\lim_{n \rightarrow \infty} \int_0^1 x^k \varphi_n(x) \, dx$$

exists for every $k = 0, 1, \dots$. Show that the limit

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) \varphi_n(x) \, dx$$

exists for every continuous function f on $[0, 1]$.

Problem 6 (Fa82). Let T be a linear transformation on a finite dimensional $\mathbb{C}$ -vector space V , and let f be a polynomial with coefficients in $\mathbb{C}$. If λ is an eigenvalue of T , show that $f(\lambda)$ is an eigenvalue of $f(T)$. Is every eigenvalue of $f(T)$ necessarily obtained in this way?

Problem 7 (Fa82). Evaluate $\int_{-\infty}^{\infty} \frac{\cos \pi x}{4x^2 - 1} \, dx$.

Problem 8 (Fa82, Fa92). Let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \mid a, b \in \mathbb{R}, a > 0 \right\} \quad \text{and} \quad N = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in \mathbb{R} \right\}$$

1. Show that N is a normal subgroup of G and prove that G/N is isomorphic to $\mathbb{R}$.
2. Find a normal subgroup N' of G satisfying $N \subset N' \subset G$ (where the inclusions are proper), or prove that there is no such subgroup.

Problem 9 (Fa82, Fa96). Let f be a real valued continuous nonnegative function on $[0, 1]$ such that

$$f(t)^2 \leq 1 + 2 \int_0^t f(s) \, ds$$

for $t \in [0, 1]$. Show that $f(t) \leq 1 + t$ for $t \in [0, 1]$.

Problem 10 (Fa82). Let a and b be complex numbers whose real parts are negative or 0. Prove the inequality $|e^a - e^b| \leq |a - b|$.

Problem 11 (Fa82, Sp06). 1. Prove that there is no continuous map from the closed interval $[0, 1]$ onto the open interval $(0, 1)$.

2. Find a continuous surjective map from the open interval $(0, 1)$ onto the closed interval $[0, 1]$.

3. Prove that no map in Part 2 can be bijective.

Problem 12 (Su79, Fa82). Let A and B be $n \times n$ matrices over a field $\mathbb{F}$ such that $A^2 = A$ and $B^2 = B$. Suppose that A and B have the same rank. Prove that A and B are similar.

Problem 13 (Fa82). Let $f_1, f_2, \dots$ be continuous functions on $[0, 1]$ satisfying $f_1 \geq f_2 \geq \dots$ and such that $\lim_{n \rightarrow \infty} f_n(x) = 0$ for each x . Must the sequence $\{f_n\}$ converge to 0 uniformly on $[0, 1]$?

Problem 14 (Fa82). Let A be an $n \times n$ complex matrix, and let A^* its conjugate transpose. Suppose that A and A^* commute with each other. Consider the linear transformations on $\mathbb{C}^n$ defined by A and A^* . Prove that they have the same image and the same kernel.

Problem 15 (Fa82). Let A be a subgroup of an abelian group B . Assume that A is a direct summand of B , i.e., there exists a subgroup X of B such that $A \cap X = 0$ and such that $B = X \oplus A$. Suppose that C is a subgroup of B and satisfying $A \subset C \subset B$. Is A necessarily a direct summand of C ?

Problem 16 (Su79, Fa79, Fa82, Su85). Find all pairs of C^∞ functions $x(t)$ and $y(t)$ on $\mathbb{R}$ satisfying

$$x'(t) = 2x(t) - y(t) \quad y'(t) = x(t)$$

Problem 17 (Fa82, Sp92, Fa16). Evaluate $\int_{-\infty}^{\infty} \frac{\sin^3 x}{x^3} dx$.

Problem 18 (Fa82). Let K be a continuous function on the unit square $0 \leq x, y \leq 1$ satisfying $|K(x, y)| < 1$ for all x and y . Show that there is a continuous function $f(x)$ on $[0, 1]$ such that we have

$$f(x) + \int_0^1 K(x, y)f(y) dy = e^{x^2}$$

Can there be more than one such function f ?

Problem 19 (Fa82). Let a and b be nonzero complex numbers and $f(z) = az + bz^{-1}$. Determine the image under f of the unit circle $\{z \mid |z| = 1\}$.

Problem 20 (Fa79, Fa82). Let G be the abelian group defined by generators x , y , and z , and relations

$$15x + 3y = 0$$

$$3x + 7y + 4z = 0$$

$$18x + 14y + 8z = 0$$

1. Express G as a direct product of two cyclic groups.
2. Express G as a direct product of cyclic groups of prime power order.
3. How many elements of G have order 2?